VISVESVARAYA TECHNOLOGICAL UNIVERSITY

JNANASANGAMA, BELGAVI-590018, KARNATAKA



MANUAL

TECHNICAL WRITING USING LATEX

(As per CBCS Scheme 2022)

Sub Name: TECHNICAL WRITING USING LATEX

Sub Code: BCSL456D

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DEPT.OF Information SCIENCE & ENGINEERING

CONTENTS

1. Develop a LaTeX script to create a simple document that consists of 2 sections [Section1, Section2], and a paragraph with dummy text in each section. And also include header [title of document] and footer [institute name, page number] in the document.

2. Develop a LaTeX script to create a document that displays the sample Abstract/Summary.

3. Develop a LaTeX script to create a simple title page of the VTU project [Use suitable Logos and text formatting].

4. Develop a LaTeX script to create the Certificate Page of the Report [Use suitable commands to leave the blank spaces for user entry].

5. Develop a LaTeX script to create a document that contains the following table with proper labels.

6. Develop a LaTeX script to include the side-by-side graphics/pictures/figures in the document by using the subgraph concept.

7. Develop a LaTeX script to create a document that consists of the following two mathematical equations.

8. Develop a LaTeX script to demonstrate the presentation of Numbered theorems, definitions, corollaries, and lemmas in the document.

9. Develop a LaTeX script to create a document that consists of two paragraphs with a minimum of 10 citations in it and display the reference in the section.

10. Develop a LaTeX script to design a simple tree diagram or hierarchical structure in the document with appropriate labels using the Tikz library.

11. Develop a LaTeX script to present an algorithm in the document using algorithm/ algorithmic/ algorithm2e Library.

12. Develop a LaTeX script to create a simple report and article by using suitable commands and formats of user choice.

Program 1

1. Develop a LaTeX script to create a simple document that consists of 2 sections [Section1, Section2], and a paragraph with dummy text in each section. And also include header [title of document] and footer [institute name, page number] in the document.

```
\documentclass{article}

% Packages
\usepackage{fancyhdr} % For header and footer
\usepackage{lipsum} % For dummy text

% Header and footer settings
\pagestyle{fancy}
\fancyhf{}
\rfoot{\thepage}
\lhead{\textit{First Program using LaTex}}
\lfoot{\textit{M.S Engineering College}}

% Document
\begin{document}

    \section{Section 1}
    \lipsum[1] % Dummy text

    \section{Section 2}
    \lipsum[2] % Dummy text

\end{document}
```

Program 2

2. Develop a LaTeX script to create a document that displays the sample Abstract/Summary.

```
\documentclass{article}
\usepackage{lipsum}

\title{Sample Abstract/Summary}
\author{}
\date{}

\begin{document}

        \maketitle

        \section*{Abstract}
        \lipsum[1]

        \vspace{0.5cm}

    \lipsum[3]

\end{document}
```

Program 3

3. Develop a LaTeX script to create a simple title page of the VTU project [Use suitable Logos and text formatting].

```
\documentclass[12pt, a4paper]{report} % Document class with font size 12pt and paper size A4
\usepackage{graphicx} % Package to include images
\usepackage{geometry} % Package to customize page layout
\geometry{a4paper, total={170mm, 257mm}, left=20mm, right=20mm, top=30mm, bottom=30mm} % Customizing page margins
\thispagestyle{empty} % Suppressing page numbers for the title page

\begin{document}

    \begin{titlepage} % Starting the title page environment
        \begin{center} % Centering content

            % University Details
            \textbf{{\large VISVESVARAYA TECHNOLOGICAL UNIVERSITY}}\\
            {\normalsize Jnana Sangama, Belgaum-590018}\\
            \vspace{0.3in}
            \includegraphics[scale=0.2]{vtu-logo.png}\\
            \vspace{0.3in}

            % Title and Project Details
            \textbf{A PROJECT REPORT} \\
            ON \\
            \vspace{0.2in}
            \textbf{{\large "Create Report Format Using LaTeX"}}\\
            \vspace{0.1in}
            {\small Submitted in partial fulfillment of the requirements for the Fifth Semester degree of
        Bachelor of Engineering in Computer Science Engineering of Visvesvaraya Technological University, Belagavi}\\
            \vspace{0.1in}
            \textbf{BACHELOR OF ENGINEERING\\IN\\COMPUTER SCIENCE AND ENGINEERING}\\
            \vspace{0.2in}

            % Student Details
            Submitted by\\
            \vspace{0.08in}
            \begin{tabular}{ll}
                \textbf{1ME21CS017} & \textbf{Braham Kumar Sah}\\
                \textbf{1ME21AI015} & \textbf{Shoaib Akhtar}\\
                \textbf{1ME21CS005} & \textbf{Aman Kumar}\\
                \textbf{1ME21CS015} & \textbf{Bikash Kumar Singh}\\
            \end{tabular}
```

```
                \vspace{0.2in}

                % Guide Details
                \textbf{Under the Guidance of}\\
                Mrs. Mala\\
                Asst. Professor Department of CSE\\

                \vspace{0.2in}

                % College Details
                \includegraphics[scale=0.4]{msec.jpg}\\
                \vspace{0.01in}
                {\small DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING}\\
                \vspace{0.1in}
                \textbf{MS ENGINEERING COLLEGE}\\
                {\small NAAC accredited Approved by AICTE, New Delhi, Affiliated to VTU,
Belagavi, Navarathna
        Agrahara, Off International Airport Road, Bengaluru -562110}\\
                \vspace{0.1in}
                {\small 2023-2024}\\

            \end{center} % Ending the center environment
        \end{titlepage} % Ending the title page environment

\end{document}
```

Program 4

4. Develop a LaTeX script to create the Certificate Page of the Report [Use suitable commands to leave the blank spaces for user entry]

```
\documentclass[12pt, a4paper]{report}
\usepackage{graphicx}
\usepackage{geometry}
\geometry{a4paper, total={170mm, 257mm}, left=20mm, right=20mm, top=20mm, bottom=20mm}
\thispagestyle{empty}

\begin{document}

	\begin{titlepage}
		\begin{center}
			\textbf{{\large VISVESVARAYA TECHNOLOGICAL UNIVERSITY}}\\
			{\normalsize Jnana Sangama, Belgaum-590018}\\
			\vspace{0.5in}
			\includegraphics[scale=0.1]{vtu-logo.png}\\
			\vspace{0.5in}
			\textbf{CERTIFICATE}\\
			\vspace{0.3in}
			This is to certify that \\
			\vspace{0.2in}
			\underline{\hspace{10cm}}\\
			\vspace{0.2in}
			(Name of the student)\\
			\vspace{0.2in}
			bearing University Seat Number \\
			\vspace{0.2in}
			\underline{\hspace{10cm}}\\
			\vspace{0.2in}
			has satisfactorily completed the project work entitled \\
			\vspace{0.2in}
			\underline{\hspace{10cm}}\\
			\vspace{0.2in}
			(Title of the project)\\
			\vspace{0.2in}
			towards the partial fulfillment of the requirements for the award of the degree of\\
			\vspace{0.2in}
			\textbf{BACHELOR OF ENGINEERING\\IN\\COMPUTER SCIENCE AND ENGINEERING}\\
			\vspace{0.5in}
			\textbf{Guide} \hspace{3.5in} \textbf{Head of the Department}\\
			\vspace{0.3in}
			\underline{\hspace{6cm}} \hspace{1.5in} \underline{\hspace{6cm}}\\
			\vspace{0.1in}
			\textbf{(Guide's Name)} \hspace{2.7in} \textbf{(HOD's Name)}\\
```

```
                \vspace{0.1in}
                \textbf{(Guide's Designation)} \hspace{2in} \textbf{(HOD's Designation)}\\
                \vspace{0.1in}
                \textbf{(Department of CSE)} \hspace{2.1in} \textbf{(Department of CSE)}\\
                \vspace{0.1in}
                \textbf{[College Name]}\\
                \vspace{0.1in}
                \textbf{[Location]}\\
                \vspace{0.1in}
                \textbf{[Month Year]} % Replace with current month and year
            \end{center}
        \end{titlepage}

\end{document}
```

Program 5

5. Develop a LaTeX script to create a document that contains the following table with proper labels.

S.No	USN	Student Name	Marks		
			Subject1	Subject2	Subject3
1	4XX22XX001	Name 1	89	60	90
2	4XX22XX002	Name 2	78	45	98
3	4XX22XX003	Name 3	67	55	59

```
\documentclass{article}
\usepackage{array, booktabs, multicol, multirow} % Load necessary packages
\renewcommand{\arraystretch}{1.2} % Adjust vertical spacing in tables

\begin{document}
	\centering
	\textbf{\Large{Student Details and Marks}} % Title
	\vspace{0.1in}

	\begin{table}[h]
		\centering
		\begin{tabular}{|c|c|c|c|c|c|} % Define table with 6 columns, all centered
			\hline
			\multirow{2}{*}{\textbf{S.No}} & \multirow{2}{*}{\textbf{USN}} & \multirow{2}{*}{\textbf{Student Name}} & \multicolumn{3}{c|}{\textbf{Marks}} \\ % Multirow for headers spanning 2 rows, Multicolumn for header "Marks" spanning 3 columns
			\cline{4-6} % Horizontal line from column 4 to 6
			& & & \textbf{Subject1} & \textbf{Subject2} & \textbf{Subject3} \\ % Sub-headers for marks
			\hline
			\multicolumn{1}{|c|}{1} & \multicolumn{1}{c|}{4XX22XX01} & \multicolumn{1}{c|}{Name 1} & 89 & 60 & 90 \\ % Data rows
			\hline
			\multicolumn{1}{|c|}{2} & \multicolumn{1}{c|}{4XX22XX02} & \multicolumn{1}{c|}{Name 2} & 78 & 45 & 98 \\
			\hline
			\multicolumn{1}{|c|}{3} & \multicolumn{1}{c|}{4XX22XX03} & \multicolumn{1}{c|}{Name 3} & 67 & 55 & 59 \\
			\hline
		\end{tabular}
	\end{table}

\end{document}
```

Program 6

6. Develop a LaTeX script to include the side-by-side graphics/pictures/figures in the document by using the subgraph concept.

```
\documentclass{article}
\usepackage{graphicx}  % Required for including images
\usepackage{subcaption}  % Required for subfigures

\begin{document}

    \begin{figure}
        \centering
        \begin{subfigure}{0.46\linewidth}  % Subfigure environment for the first image
            \includegraphics[width=\linewidth]{image1.jpg}  % Include image1.jpg
            \caption{Caption for image 1}  % Caption for the first image
            \label{fig:subfig1}  % Label for referencing the first image
        \end{subfigure}
        \hfill  % Add horizontal space between subfigures
        \begin{subfigure}{0.44\linewidth}  % Subfigure environment for the second image
            \includegraphics[width=\linewidth]{image2.png}  % Include image2.png
            \caption{Caption for image 2}  % Caption for the second image
            \label{fig:subfig2}  % Label for referencing the second image
        \end{subfigure}
        \caption{Combined caption for both images}  % Overall caption for the figure
        \label{fig:subfigures}  % Label for referencing the entire figure
    \end{figure}

\end{document}
```

Program 7

7. Develop a LaTeX script to create a document that consists of the following two mathematical equations.

$$x = \frac{-b \pm \sqrt{b^2-4ac}}{2a}$$

$$= \frac{-2 \pm \sqrt{2^2-4*(1)*(-8)}}{2*1}$$

$$= \frac{-2 \pm \sqrt{4+32}}{2}$$

$$\varphi_\sigma^\lambda A_t = \sum_{\pi \in C_t} \operatorname{sgn}(\pi) \varphi_\sigma^\lambda \varphi_\pi^\lambda$$

$$= \sum_{\tau \in C_{\sigma t}} \operatorname{sgn}(\sigma^{-1}\tau\sigma) \varphi_\sigma^\lambda \varphi_{\sigma^{-1}\tau\sigma}^\lambda$$

$$= A_{\sigma t} \varphi_\sigma^\lambda$$

```
\documentclass{article}
\usepackage{amsmath} % Required for mathematical environments and commands

\begin{document}

    \section*{Equations Set 1} % Section header for the first set of equations

    \begin{align} % Begin the align environment for multiple equations
        x &= -b \pm \sqrt{\sqrt[4]{b^2 - 4ac}} \notag \\ % First equation with \notag to suppress numbering
        x &= \frac{-b \pm \sqrt{22 - 4 \cdot 1 \cdot (-8)}}{2 \cdot 1} = \frac{-b \pm \sqrt{\sqrt{4 + 32}}}{2} = \frac{-b \pm \sqrt{2}}{2} \notag % Second equation with \notag to suppress numbering
    \end{align} % End the align environment

    \section*{Equations Set 2} % Section header for the second set of equations

    \begin{align} % Begin the align environment for multiple equations
        \varphi_{\sigma}^{\lambda} \cdot A_{t} &= \sum_{\pi \in C_{t}} \text{sgn}(\pi) \cdot \varphi_{\sigma}^{\lambda} \cdot \varphi_{\pi}^{\lambda} \notag \\ % First equation with \notag to suppress numbering
        &= \sum_{\tau \in C_{\sigma}^{t}} \text{sgn}(\sigma^{-1} \tau \sigma) \varphi_{\sigma}^{\lambda} \varphi_{\sigma^{-1} \tau \sigma}^{\lambda} \notag \\ % Second equation with \notag to suppress numbering
        &= A_{\sigma}^{t} \varphi_{\sigma}^{\lambda} \notag % Third equation with \notag to suppress numbering
    \end{align} % End the align environment

\end{document}
```

Program 8

8. Develop a LaTeX script to demonstrate the presentation of Numbered theorems, definitions, corollaries, and lemmas in the document.

```
\documentclass{article}
\usepackage{amsthm}

% Define theorem-like environments
\newtheorem{theorem}{Theorem}[section] % Theorems numbered within sections
\newtheorem{definition}[theorem]{Definition} % Definitions share numbering with theorems
\newtheorem{corollary}[theorem]{Corollary} % Corollaries share numbering with theorems
\newtheorem{lemma}[theorem]{Lemma} % Lemmas share numbering with theorems

\begin{document}

	\section{Introduction}

	\begin{theorem}
		This is a theorem.
	\end{theorem}

	\begin{definition}
		This is a definition.
	\end{definition}

	\begin{corollary}
		This is a corollary.
	\end{corollary}

	\begin{lemma}
		This is a lemma.
	\end{lemma}

	\section{Another Section}

	\begin{theorem}
		Another theorem.
	\end{theorem}

\end{document}
```

Program 9

9. Develop a LaTeX script to create a document that consists of two paragraphs with a minimum of 10 citations in it and display the reference in the section.

Note: First create any_name.tex text file and below code copy and paste it in that file.

```
\documentclass{article}
\usepackage{cite}

\begin{document}

    \title{Sample Document with Citations}
    \author{}
    \date{}
    \maketitle

    \section{Emerging Powers in International Politics}
     he 21st century is marked by an increased attention to the appeal and positive image of a country as instruments of influence in the international arena\cite{bohomolov2012ghost}. There has appeared the concept of soft power, whose author, U.S\cite{sergunin2015understanding}. political scientist Joseph Nye described it as “the ability to get what you want through attraction rather than coercion or payments\cite{hill2006moscow}.” A nation’s image secures attractiveness and trust in a country, playing a crucial role as the key soft power component\cite{kiseleva2015russia}. Therefore, the efforts of states along this line relate not so much to the sphere of culture and information as to geopolitics\cite{kosachev2012spsecific}.


    \section{Atomic Force Microscopy, a Powerful Tool in Microbiology}
  Understanding the functions of microbial cell surfaces requires knowledge of their structural and physical properties\cite{dufrene2002atomic}. Electron microscopy has long been recognized as a key technique in microbiology to elucidate cell surface ultra structure\cite{engel1999atomic}. An exciting achievement has been the development of cryotechniques which allow high-resolution imaging of cell structures in conditions close to the native state\cite{franz2008atomic}. Yet direct observation in aqueous solution remained impossible.Because of the small size of microorganisms, the physical properties of their surfaces have been difficult to study\cite{marrese2017atomic}. Quantitative and qualitative information on physical properties can be obtained by electron microscopy techniques, X-ray photoelectron spectroscopy, infrared spectroscopy, contact angle, and electrophoretic mobility measurements\cite{altman2015noncontact}.

    \bibliographystyle{plain}
    \bibliography{references}

\end{document}
```

Note: Again create one more references.bib text file and below code copy and paste it in that text file.

```
@article{kosachev2012spsecific,
	title={The specifics of Russian soft power},
	author={Kosachev, Konstantin},
	journal={Russia in Global Affairs},
	volume={7},
	number={3},
	pages={1--11},
	year={2012},
	publisher={Фонд исследований мировой политики}
}

@article{sergunin2015understanding,
	title={Understanding Russia's soft power strategy},
	author={Sergunin, Alexander and Karabeshkin, Leonid},
	journal={Politics},
	volume={35},
	number={3-4},
	pages={347--363},
	year={2015},
	publisher={SAGE Publications Sage UK: London, England}
}

@article{kiseleva2015russia,
	title={Russia's soft power discourse: identity, status and the attraction of power},
	author={Kiseleva, Yulia},
	journal={Politics},
	volume={35},
	number={3-4},
	pages={316--329},
	year={2015},
	publisher={SAGE Publications Sage UK: London, England}
}

@book{bohomolov2012ghost,
	title={A ghost in the mirror: Russian soft power in Ukraine},
	author={Bohomolov, Oleksandr and Lytvynenko, Oleksandr Valeri{\u\i}}ovych},
  year={2012},
  publisher={Chatham House London}
}

@article{hill2006moscow,
	title={Moscow discovers soft power},
	author={Hill, Fiona},
	journal={Current History},
	volume={105},
	number={693},
	pages={341--347},
```

year={2006},
publisher={University of California Press}
}

@article{dufrene2002atomic,
title={Atomic force microscopy, a powerful tool in microbiology},
author={Dufr{\^e}ne, Yves F},
journal={Journal of bacteriology},
volume={184},
number={19},
pages={5205--5213},
year={2002},
publisher={Am Soc Microbiol}
}

@article{engel1999atomic,
title={Atomic force microscopy: a powerful tool to observe biomolecules at work},
author={Engel, Andreas and Lyubchenko, Yuri and M{\"u}ller, Daniel},
journal={Trends in cell biology},
volume={9},
number={2},
pages={77--80},
year={1999},
publisher={Elsevier}
}

@article{franz2008atomic,
title={Atomic force microscopy: a versatile tool for studying cell morphology, adhesion and mechanics},
author={Franz, CM and Puech, P-H},
journal={Cellular and Molecular Bioengineering},
volume={1},
pages={289--300},
year={2008},
publisher={Springer}
}

@article{marrese2017atomic,
title={Atomic force microscopy: a powerful tool to address scaffold design in tissue engineering},
author={Marrese, Marica and Guarino, Vincenzo and Ambrosio, Luigi},
journal={Journal of functional biomaterials},
volume={8},
number={1},
pages={7},
year={2017},
publisher={MDPI}
}

@article{altman2015noncontact,

title={Noncontact atomic force microscopy: an emerging tool for fundamental catalysis research},
author={Altman, Eric I and Baykara, Mehmet Z and Schwarz, Udo D},
journal={Accounts of Chemical Research},
volume={48},
number={9},
pages={2640--2648},
year={2015},
publisher={ACS Publications}
}

Program 10

10. Develop a LaTeX script to design a simple tree diagram or hierarchical structure in the document with appropriate labels using the Tikz library.

```
\documentclass{article}
\usepackage{tikz}

\begin{document}
	\centering

	% Define styles for nodes
	\tikzstyle{level 1}=[level distance=4cm, sibling distance=6cm]
	\tikzstyle{level 2}=[level distance=4cm, sibling distance=3cm]

	% Begin TikZ picture
	\begin{tikzpicture}[grow=down, sloped]
		% Root node
		\node {Root}
		% First child
		child {
			node {Child 1} % First child node
			child {
				node {Subchild 1} % Subchild node
			}
			child {
				node {Subchild 2} % Subchild node
			}
		}
		% Second child
		child {
			node {Child 2} % Second child node
			child {
				node {Subchild 1} % Subchild node
			}
			child {
				node {Subchild 2} % Subchild node
			}
		};
	\end{tikzpicture}

\end{document}
```

Program 11

11. Develop a LaTeX script to present an algorithm in the document using algorithm/ algorithmic/ algorithm2e Library.

```
\documentclass{article}

\usepackage{algorithm}
\usepackage{algpseudocode}

\begin{document}

	\begin{algorithm}
		\caption{Bubble Sort}
		\begin{algorithmic}[1]
			\Procedure{BubbleSort}{$A, n$}
			\For{$i \gets 0$ to $n-1$}
			\For{$j \gets 0$ to $n-1-i$}
			\If{$A[j] > A[j+1]$}
			\State Swap $A[j]$ and $A[j+1]$
			\EndIf
			\EndFor
			\EndFor
			\EndProcedure
		\end{algorithmic}
	\end{algorithm}

\end{document}
```

Program 12

12. Develop a LaTeX script to create a simple report and article by using suitable commands and formats of user choice.

```
\documentclass{report}

\title{Simple Report}
\author{}
\date{}

\begin{document}

    \maketitle

    \tableofcontents

    \chapter{Introduction}
    This is the introduction.

    \chapter{Methods}
    This is the methods section.

    \chapter{Results}
    This is the results section.

    \chapter{Discussion}
    This is the discussion section.

    \chapter{Conclusion}
    This is the conclusion.

\end{document}
```